

Enhancing Memory Recall Through AI-Assisted Method of Loci in Virtual Reality

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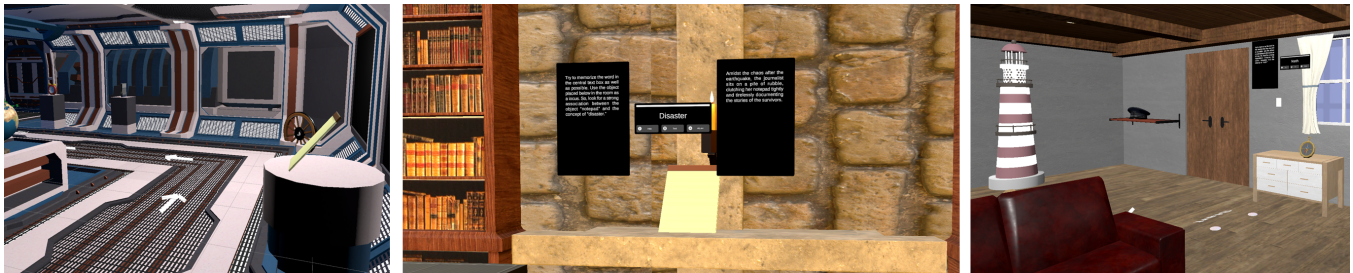


Figure 1: The virtual sci-fi environment (left), an example of an AI-generated text for an associated object (middle) and the cabin environment with a matched Object-word pair (right).

Abstract

The Method of Loci is a well-established mnemonic technique that involves associating words with objects placed along a route. A key factor in its effectiveness is creating meaningful connections between the words to-be-remembered and the corresponding objects. In this study, we investigate how artificial intelligence (AI) can enhance this technique by i) selecting appropriate objects for each word and ii) generating coherent textual associations between the words and their objects. These AI-assisted approaches are compared to a control condition, where iii) object-word pairs are chosen randomly without assistance. Our findings demonstrate that the *Object* condition significantly improves word recall both immediately and after one week. On the other side, the *Text* condition did not lead to a significant enhancement in recall, and perceived workload showed no significant differences across all conditions. These results offer valuable insights for advancing mnemonic techniques and suggest directions for future research to optimize memory strategies.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Empirical studies in HCI*.

Keywords

Artificial Intelligence, Mnemonic Techniques, Virtual Reality, Method of Loci, Memory

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1 Introduction

The world's most successful memory athletes demonstrate memory performances that seem almost impossible. For example, the world record for learning random word lists in 15 minutes, set by Emma Alam, is 410 correctly recalled words [7]. Most Super Memorizer utilize mnemonic techniques, which are learning strategies that help retrieve learned information by associating it with more meaningful information [16].

One of the most effective mnemonic techniques is the Method of Loci [15], which leads to improved memory performance [3, 14, 15, 19]. It is based on associating information to locations (loci) along a route, which can later be retrieved by mentally traversing that route [21]. In recent years, digital implementations have gained attention. Legge et al. [11] asked participants to apply the Method of Loci in a virtual 3D environment displayed on a computer monitor, traditionally, and without a strategy, with comparable results for the two Loci groups. Similarly, Fassbender and Heiden [5] found improved memory performance compared to a control group without a learning strategy when participants used a virtual environment presented on a monitor. The Method of Loci has also been explored in virtual reality (VR) [6, 9, 12, 13, 18]. Here, the



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comparison between applying the Method of Loci on a monitor versus in a head-mounted display (HMD) showed better results in the HMD group, although not significant [9], or comparable results [13]. Compared to pen and paper, higher recall scores were measured in a VR study that also employed color-coded information to improve associations [6]. Virtual environments offer a number of potential advantages for applying the Method of Loci. For example, they can represent any number of rooms, and users are not dependent on their natural surroundings. Unlike variants that display space on a monitor, VR preserves three-dimensionality, which could benefit spatial memory. In particular, VR environments provide the opportunity to independently design the environment. Through generative AI, this design can be automated, potentially allowing the learning environment to be tailored to the learning material with little effort.

Building on prior research showing that stronger cue-recall connections improve memory performance [17], this study introduces a novel approach: utilizing AI to facilitate the association of words to their corresponding objects within VR. This aims to create highly meaningful, individualized associations between objects and learned information, thereby strengthening the spatial memory cues at the core of the Method of Loci. By dynamically aligning objects to specific words through AI-generated, contextually relevant pairings, we seek to optimize the creation of robust, distinctive links that significantly enhance recall.

Consequently, this research focuses on how AI-driven word-object matching in VR can improve the effectiveness and efficiency of the Method of Loci. For this, we compared three different approaches: i) matching the word to be learned to a fitting *Object* using AI, ii) providing an AI-generated *Text* that describes a scenario linking an object and a word together, helping to establish a connection between the two, and iii) a control condition *Without AI Assistance*, where a word is randomly assigned to an object. We assessed the perceived workload, and the amount of remembered words - both directly after the study, and one week later - and our results provide important implications for future memory training in immersive environments.

Our hypotheses were the following:

- **H1:** The AI-assisted conditions lead to remembering more words directly after the learning task than the condition without AI assistance, both for the AI-chosen objects (H1.a) and the AI-generated texts (H1.b).
- **H2:** The AI-assisted conditions lead to remembering more words one week after the learning task than the condition without AI assistance, both for the AI-chosen objects (H2.a) and the AI-generated texts (H2.b).
- **H3:** The AI-assisted conditions lead to a lower workload than the condition without AI assistance, both for the AI-chosen objects (H3.a) and the AI-generated texts (H3.b).

2 Methods

We conducted a within-subject user study to investigate the effect of AI-assistance on recall effectiveness and perceived workload of the Method of Loci in VR. The idea of this work is to examine how AI-based support can improve learning and reduce perceived workload, especially the mental demand. The success of mnemonic

cueing depends on how well a strong connection between the word and the object can be established [17]. Therefore, both support approaches focus on improving the connection between the word and the object.

2.1 Implementation

For the study, an application was developed for the Meta Quest 3 using the Unity engine version 2022.3. The application featured three virtual rooms, each designed for learning 20 words. Although the Method of Loci remains an effective technique when using the same environment multiple times, retroactive interference effects [1] may occur. To ensure that multiple learning sessions did not lead to confusion or information overload, three different rooms were used, each visually and structurally distinct: A sci-fi room, a medieval room and a furnished cabin. For each word to be learned, an object was placed in the room to serve as a locus associated with that word. The word to be learned was displayed in a UI window above the object during this process (See Figure 1). Participants could move around the room using teleportation hotspots.

In each condition, the participants learned a list of 20 words. The room and condition varied each time, ensuring that all participants experienced all conditions in a counterbalanced order to control for learning effects. A total of 60 words were used across all participants, with the order randomized. These were all German nouns selected randomly [4] (see supplementary material for the words). We used a pool of 60 objects, all of which were assigned to a word according to one of the following three conditions: The *Object* condition, the *Text* condition and the condition *Without AI Assistance*.

In the *Object* condition, the objects placed in the room are adjusted to the words to be learned so that their association is as intuitive as possible. The selection of an object for each word was performed using Chat GPT version 4.1. Since real-time AI-based 3D object generation was not yet reliably available during our study, this part of the application was implemented as a Wizard of Oz prototype, meaning that word-object pairs were pre-chosen prior to the study. For each AI-selected object, a suitable 3D mesh was either manually chosen from templates or created independently before the start of the study. The following prompt was used for object selection:

You should support me in designing an environment for applying the Method of Loci. I will tell you a word that should be associated with the Method of Loci. Make ten suggestions for an object that can serve as a loci for learning the word, especially chosen because it is particularly easy to establish a connection to the word. Please check each object to see if it is a single object. If not, please remove it from the list. Compound nouns are allowed as long as they define a clear new concept. Remove only objects that cannot be clearly defined as a single object. Also, exclude words that are synonyms for the word. Additionally, exclude words that are just a specification of the main word or a representational form. For each

suggestion, evaluate it from 0 to 100 based on the criteria: specificity, visualizability, and how realistically the object can be placed in a room. The object with the highest score will be selected.

In the *Text* condition, support is provided by presenting a text within the VR environment that describes a scenario linking the word and the object together. The text can be used directly as a connection or as inspiration for creating one's own association. Here, too, the goal is to reduce cognitive load. By integrating a direct connection to the Chat GPT API from within the application, real-time generation of the text was enabled. During use, the following prompt was supplemented with relevant words and objects:

Analyze the properties, functions, and common cultural or factual contexts of [object] and [word]. Find a clear, plausible connection rooted in reality or well-known contexts (e.g., technical use, typical scenarios, popular stories, or terms). Describe in one or two clear sentences a vivid scene in which [object] and [word] are inextricably and memorably linked through this connection. Allow additional elements only if they directly enhance this link. Avoid unnecessary embellishments or bizarre details. Provide only the scene description, without introduction or explanation.

As a control condition, a condition *Without AI Assistance* was used in which none of the supports could be utilized. In this scenario, words and objects were randomly paired from the object and word pool.

2.2 Procedure

At the beginning of the study, participants were informed about its purpose and signed an informed consent form. Then, they provided demographic information and details about their prior experience with VR and the Method of Loci.

Once they entered VR, participants completed an interactive tutorial explaining controls, navigation, the task, and the UI. They practiced button operations and teleportation within the tutorial room, guided by an audio tutorial and support from the experimenter if needed. Then, they experienced the first condition in a counterbalanced order by activating a start button. They then navigated from object to object until all 20 objects along the route were visited. Afterward, they removed the headset to complete the NASA-TLX questionnaire [8] on a scale from 0 to 100, with particular interest in mental demand since the AI-supported features were expected to reduce mental load by making it easier to find associations between words and objects (**H3**). Next, they were instructed to recall and write down as many words as possible based on the objects they remembered (**H1**). After all three conditions, they answered further questions about the entire study, e.g., how many of the AI-generated texts they read, what they perceived as helpful, difficult, or they were able to leave general comments.

One week after the in-person session, participants completed an online follow-up questionnaire, where they were asked to recall the learned words again (**H2**). This was meant to test not only for immediate effects, but also for lasting ones. Further, we wanted to control for ceiling effects that might occur in the immediate recall.

3 Results

Thirty-six participants took part in the study, with a mean age of 27.17 ± 6.98 years. Eighteen participants identified as female, and 18 as male, and most of them were students participating for course credit. All of them could speak German. For the word recall after one week, two participants did not submit a response, therefore, only data for 34 participants is available here (17 male, 17 female).

We evaluated the results using R, by first testing them for normal distribution using a Shapiro-Wilk test. Since data was not normally distributed, we used a Friedman test for identifying whether there are differences between the condition, with post-hoc Wilcoxon Signed-Rank tests with Holm correction in case of significant results. A significance level of 0.05 was used for all tests. All descriptive results can be found in Table 1.

Measurement	w/o AI Assistance	Object	Text	p-value
Immediate Word Recall	14.68 (3.81)	16.31 (3.21)	14.18 (4.31)	0.0088
One Week Word Recall	4.92 (3.36)	7.38 (4.75)	6.22 (5.18)	0.005
Mental Demand	53.05 (21.72)	44.44 (22.67)	54.03 (22.07)	0.0697
Physical Demand	22.22 (23.83)	19.58 (19.83)	23.61 (23.71)	0.1211
Temporal Demand	19.17 (21.53)	21.25 (18.57)	21.67 (21.55)	0.7563
Performance	52.5 (26.47)	44.86 (30.62)	52.64 (25.03)	0.4253
Effort	48.19 (21.98)	43.19 (22.87)	50.28 (24.26)	0.132
Frustration	26.53 (20.76)	17.78 (15.79)	26.94 (22.94)	0.0027

Table 1: Descriptive results (mean and SD) of the word recall tasks, as well as NASA-TLX results, and p-value for the Friedman test for the conditions *Without AI Assistance*, *Object*, and *Text*.

3.1 Word Recall

For immediate word recall, a Friedman test resulted in a p-value of 0.0088. Post-hoc Wilcoxon Signed Rank tests with Holm correction showed a significant difference between the *Object* condition and the condition *Without AI Assistance* ($p = 0.04$), and for the *Object* condition and the *Text* condition ($p = 0.001$), while the other comparison was not significant (*Without AI Assistance* vs. *Text*: $p = 0.399$).

One week after the VR intervention, the participants were asked to recall the words again. A Friedman test showed a significant difference ($p = 0.005$), and a post-hoc test with Holm correction indicated that more words were remembered with the *Object* condition, compared to the condition *Without AI Assistance* ($p = 0.0041$). The other two tests showed no significant differences (*Without AI Assistance* vs *Text*: $p = 0.1858$, *Object* vs *Text*: $p = 0.1858$).

3.2 Perceived Workload

For the NASA-TLX questionnaire, only the Frustration subscale showed a statistically significant difference when evaluated using the Friedman test ($p = 0.0027$). However, subsequent post hoc pairwise comparisons with Wilcoxon Signed-Rank tests, adjusted with

Holm correction, did not yield any significant pairwise differences ($p = 0.17$ (*Without AI Assistance* vs. *Object*), $p = 0.15$ (*Text* vs. *Object*) and $p = 0.86$ (*Without AI Assistance* vs. *Text*)). For none of the other subscales of the NASA-TLX questionnaire, a significant difference between the conditions could be found (See Table 1). Therefore, no more pairwise comparisons were performed.

4 Discussion

Overall, the study provides evidence that AI-based selection of suitable *Objects* can enhance the effectiveness of the Method of Loci. It also appears to influence long-term memory; both recall scores were significantly higher for the *Object* condition, compared to the condition *Without AI Assistance*, confirming **H1.a** and **H2.a**. For immediate recall, the scores were even significantly higher for the *Object* condition, compared to the *Text* condition. Our qualitative feedback indicated that some participants found the modified objects helpful, while others reported that less fitting objects led to more unusual or specific associations, making them more memorable. Individually pairing objects and words according to user preferences might be an area for future research, and AI could also assist to automate this approach. Additionally, interaction possibilities with objects could foster deeper processing [10] if such interactions are meaningfully integrated into the association process. This could further strengthen the AI assisted object selection.

The AI-generated *Texts* on the other side did not improve memory performance, and **H1.b** and **H2.b** could not be confirmed. The qualitative feedback was mixed; some participants found the texts helpful, but more frequently, they were described as too vague, unhelpful, confusing, or distracting. Previous research shows that personally formed associations are more effective than externally suggested ones in the context of eyewitness recall [20]. Thus, external texts can only be supportive if they are of high quality and comparable in effectiveness to self-generated connections. However, this lays the groundwork for further research. In particular, improved prompting or more complex support structures could enhance memory performance, and, again, associations could be tailored to individual users. Instead of suggesting complete associations, a dialog-based AI system could also interactively help users identify suitable associations through conversation.

The last factor we evaluated was perceived workload. Here, we could not find any significant differences, and our hypotheses **H3.a** and **H3.b** could not be confirmed. The workload overall was quite high, especially for mental demand and effort. For the *Text* condition, we assume that reading the text resulted in additional cognitive workload, which might have been distracting from the primary task of memorizing the words; but again, this was not significant. Future work could also investigate how different modalities, e.g., auditory information, influence perceived workload in forming meaningful associations between objects and words.

Limitations and Future Work

The limitations of this study primarily stem from the participant group. Since the majority were students, it is likely that they are more accustomed to memorization. As it was done in prior work [14] it would be interesting to let them memorize words or constructs relevant to their studies, or establish meaningful learning locations

for single concepts, similar to the robotic operation training for astronauts that was researched by Costantini et al. [2]. Additionally, our participants had above-average prior experience with VR headsets and familiarity with the Method of Loci, which could have influenced the results. Another limitation is that the specific 3D objects in the *Object* condition were selected before the study using a Wizard of Oz approach. Further research is needed to determine how effective fully AI-generated objects would be, which is essential for later implementation as an actual learning aid. Moreover, only one learning session was conducted here; in practical applications, participants would likely revisit the virtual space multiple times to reinforce associations. Additionally, the effectiveness of training with the Method of Loci is known to improve with practice [3], so well-trained users might rely less on VR support. Future studies are encouraged to look into these areas in more depth.

5 Conclusion

The aim of this study was to investigate the effectiveness of learning words using an AI-assisted VR implementation of the Method of Loci, compared to a VR implementation *Without AI Assistance*. Three conditions were examined in a within-subject design, focusing on how the use of AI could facilitate or strengthen the formation of connections between the learning content and cue stimuli. In the *Object* condition the objects used as loci in the virtual environment were semantically adapted by AI to the words to be learned. In the *Text* condition, AI-generated support texts were provided to help establish the association between objects and words. A control condition *Without AI Assistance* showed randomly selected object-word pairs. In a user study, 36 participants learned 20 words and were asked to recall them immediately after learning and after one week. The results show that supporting the *Object* selection with AI significantly improved the number of words remembered compared to the condition *Without AI Assistance*, both immediately and after one week. In contrast, the support texts did not lead to a significant improvement in memory performance. None of the conditions had a significant impact on perceived workload.

Future research could focus on analyzing the exact causes of these observed outcomes and exploring how the finding that AI-tailored objects can enhance word recall can be integrated into practical applications, such as training programs. Overall, it was demonstrated that AI-selected *Objects* in virtual spaces have the potential to further enhance one of the most effective known memory techniques.

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